

FINAL PROJECT — LINEAR SYSTEM

Progress Report Due: Midterm Week

Final Project Report Due: Week 16

Make a group of five and follow the instructions below. Compile a report based on your results.

1. Find a system that can be represented in a **second order differential equation** from a reputable journal (IEEE or scopus-indexed). EXAMPLE IN TEXTBOOK ARE NOT ALLOWED.
2. Provide the explanation of the system (e.g. what represent as input and output and what would the physiological relationship between them).
3. Represent the system into a second order differential equation and explain the model (what does the parameter and variables of the model represent in the real system).
4. Analyze the system to find whether it is linear, time-invariant, causal, stable, or not.
5. Explain the effect of a **single rectangular pulse** input on the system in
 - (a) time domain,
 - (b) Laplace domain, and
 - (c) frequency domain.Use the continuous-time Fourier transform for part (c).
6. Find out the Fourier series representation of the input and output from Step 5. Compare the Fourier series and the real signal to find how close the Fourier representation resembles the real signal. How many terms is good enough for the Fourier series to resemble the real signal and provide your explanation.
7. What would the transfer function of the system be in
 - (a) Laplace domain and
 - (b) frequency domain.
8. Analyze the system based on its transfer function (at least must contain s-plane and pole-zero analysis).
9. Determine impulse and step response of the system in
 - (a) time domain,
 - (b) Laplace domain, and
 - (c) frequency domain.

Report must be written using the following template:

(https://drive.google.com/file/d/1IXPUI_4-7LvEXL1RYex1aoJ01vm9Z0J/view?usp=sharing).

Include any source and code you utilize while following the instruction.

Your report will be judged based on:

- a. Complexity of the model
- b. Simplicity of the solution
- c. Elegance of the language (English and MATLAB/Python/any programming language you choose)
- d. Aesthetic of the report
- e. Presentation performance

Progress Report must contain solutions to Problem 1–4, 9(a).